



Sourabh Jain

B.Tech — Computer Science & Engineering — 2027

9499176693 — sourabh.jain.23cse@bmu.edu.in

LinkedIn — GitHub — LeetCode — Portfolio

Education

Degree	Year	Institution	Score
B.Tech, Computer Science & Engineering	2023 – 2027	BML Munjal University	CGPA: 6.4/10
Class XII (CBSE)	2021 – 2022	S.D. Public School	69%
Class X (CBSE)	2019 – 2020	S.D. Public School	89%

Relevant Coursework

Data Structures & Algorithms, Object Oriented Programming, DBMS, Operating Systems, Computer Networks, Machine Learning, NLP

Technical Skills

Languages: C++, Python, JavaScript, Java, SQL, HTML | **Libraries:** NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib, Seaborn | **Tools:** Git, GitHub, Postman, Karate DSL, Mockoon, IntelliJ IDEA CE | **Concepts:** Data Structures & Algorithms, REST APIs, API Testing, Test Automation, Feature Engineering, Regex Validation

Experience

Airtel Payments Bank (*SDE Intern – API Testing & Automation*) Jun 2025 – Jul 2025

- Tasked with strengthening QA coverage for Aadhaar, PAN, and OTP identity-verification services ahead of a release; built automated API test suites in Karate DSL to validate AUA, KUA, KUA-OTP, and OTP banking endpoints on REST architecture in IntelliJ IDEA CE.
- Needed to isolate dependent services for reliable, repeatable testing; mocked external systems with Mockoon and validated request/response contracts using Postman, enabling consistent test execution across environments.
- Faced slow, error-prone manual regression cycles; authored 40+ reusable test cases covering positive, negative, and edge-case scenarios, cutting manual testing effort by 40% and speeding up release feedback.

Projects

AI Code Review Assistant Jan 2026 – Present

- Aimed to replace slow manual code reviews with an automated, production-grade platform; built the full stack using React, Node.js, Express, and MongoDB, with Redis for caching and Gemini AI for automated code analysis.
- Needed context-aware, secure review sessions; implemented a Retrieval-Augmented Generation (RAG) pipeline with LangChain and ChromaDB alongside BullMQ background jobs and JWT authentication, delivering an end-to-end system that surfaces repository-level insights.

Plant Disease Detection using Digital Image Processing Dec 2025

- Designed an automated computer vision pipeline to classify plant leaf diseases from raw imagery; engineered advanced augmentation workflows using Albumentations and OpenCV (blurring, rotation, scaling) to enhance model generalizability on limited training sets.
- Implemented deep learning classification models and automated image preprocessing pipelines to extract localized leaf features, achieving robust multi-class disease identification for precision agriculture applications.

Gridlock AI: Decision Intelligence for Traffic Management Jan 2026

- Set out to identify urban congestion hotspots and optimize parking enforcement through data-driven decision intelligence; built a Python/FastAPI backend with a Streamlit dashboard, using DuckDB for fast analytical querying and LightGBM/XGBoost for predictive modeling.
- Needed actionable, real-time operational guidance; implemented route optimization, predictive analytics, and resource allocation models feeding interactive dashboards, producing a platform that pinpoints hotspots and supports smarter enforcement decisions.

Air Quality Index (AQI) Analysis Aug 2025 – Dec 2025

- Sought to uncover pollution trends and city-wise air quality patterns hidden in a large, unstructured dataset; processed 50K+ AQI records across 12 cities using Python and Pandas, applying thorough data preprocessing and exploratory analysis.
- Needed to make patterns interpretable for stakeholders; built Matplotlib visualizations surfacing correlations and trends across cities and pollutant types, generating actionable environmental insights from the large-scale data.

Honors & Certifications

- **1st Position – Track 1, EIS4.0 (Research & Innovation):** Ranked first among competing teams for a research and innovation project submission.
- **Python Programming Certification – Udemy (2025):** Completed a structured course covering core Python programming and problem-solving fundamentals.
- **Web Development Certification – Mood Indigo, IIT Bombay:** Completed hands-on training in front-end and back-end web development fundamentals.